

Short review of “traditional” approaches for decision and uncertainty

- Uncertainty may come from:
 - Randomness,
 - Incomplete information, or
 - Variability in how systems behave.
 - The goal is to find strategies that perform well across different possible scenario.
 - Systems thinking = toolkit for designing complex systems from the bottom up.
 - We often assume we can model, map, control the parts, predict outcomes.
 - Failures happen: technical breakdowns, runaway costs, unintended consequences.
The idea is that systems thinking gives us confidence, but real systems push back.
We need to see where decision-making under complexity often misfires when assuming static, fully known systems.
 - Some other examples of **decision-making under risk** that can be used for discussions:
Each of these examples allows calculation of expected values, discussion of risk preferences, and can be extended with utility functions.
1. **Oil drilling investment:** A company can drill for oil in a new field:
 - **Option A (Do nothing):** No risk, no reward.
 - **Option B (Drill):** 25% chance of striking oil and making \$5 million, 75% chance of losing \$1 million in drilling costs.
 2. **Weather-dependent farming:** A farmer chooses between:
 - **Option A (Wheat):** Stable profit of \$20,000 regardless of weather.
 - **Option B (Corn):** If rainfall is good (40% chance), profit is \$50,000; if drought (60% chance), loss of \$5,000.
 3. **Gambling vs. saving:** A person can either:
 - **Option A (Save):** Put \$1,000 in a bank account earning 4% interest (\$1,040 after one year).
 - **Option B (Bet):** 10% chance to win \$10,000, 90% chance to lose it all.
 4. **Job offer with bonus risk:** A graduate has two job offers:
 - **Option A (Stable firm):** Salary \$60,000 guaranteed.
 - **Option B (Startup):** Base salary \$40,000 + 30% chance of a \$60,000 bonus, but 70% chance of no bonus.
 5. **Medical treatment choice:** A patient must decide whether to undergo an experimental treatment:
 - **Option A (standard treatment):** 90% chance of living 5 more years, 10% chance of complications leading to only 2 years.
 - **Option B (experimental treatment):** 50% chance of living 15 more years, 50% chance of surviving only 1 year.

Example

6. **Bonds vs. startup:** A person has saved \$10,000 and is considering two investment options:

1. **Government bonds (safe)**

Guaranteed return of 5% at the end of the year.

2. **New startup (risky)**

50% chance of success, doubling the money; 50% chance of failure, recovering only half.

Outcomes

- **Bonds (certain):** Final wealth = $\$10,000 \times 1.05 = \$10,500$.
- **Startup (risky):**
 - Success ($p = 0.5$): \$20,000.
 - Failure ($p = 0.5$): \$5,000.

Expected values

- $EV(\text{Bonds}) = 10,500$
- $EV(\text{Startup}) = 0.5 \times 20,000 + 0.5 \times 5,000 = 10,000 + 2,500 = 12,500$

Decision-making

- **Risk-neutral:** choose the startup (higher expected value).
- **Risk-averse:** may prefer bonds despite the lower EV.
- **Risk-seeking:** more inclined toward the startup.

Questions

1. At what bond interest rate r^* would you be indifferent between bonds and the startup (i.e., when $\$10,000(1 + r^*) = 12,500$)?
2. If the startup success probability drops to 40%, how does the decision change?
3. Introduce a utility function (e.g., $U(W) = \sqrt{W}$) and compare $\mathbb{E}[U]$ for the two options.

Answers:

- Initial wealth: \$10,000.
- Bonds (certain): payoff \$10,500.
- Startup (risky): with probability p get \$20,000, with probability $1 - p$ get \$5,000.
- In the baseline, $p = 0.5$.

Q1. Indifference bond rate

Find the bond rate r^* that makes an investor indifferent between Bonds and the baseline Startup in *expected value* terms:

$$10,000(1 + r^*) = \mathbb{E}[\text{Startup}] = 0.5 \cdot 20,000 + 0.5 \cdot 5,000 = 12,500.$$

Hence

$$1 + r^* = \frac{12,500}{10,000} = 1.25 \implies \boxed{r^* = 25\%}.$$

Q2. Lower success probability $p = 0.4$

Compute the expected value of the Startup when $p = 0.4$:

$$\mathbb{E}[\text{Startup} | p = 0.4] = 0.4 \cdot 20,000 + 0.6 \cdot 5,000 = 8,000 + 3,000 = \boxed{11,000}.$$

Comparison with Bonds (certain \$10,500):

$$11,000 > 10,500 \implies \text{a risk-neutral decision-maker still prefers the Startup.}$$

Threshold for indifference by success probability. Solve for p such that $\mathbb{E}[\text{Startup}] = 10,500$:

$$20,000p + 5,000(1 - p) = 10,500 \implies 15,000p + 5,000 = 10,500 \implies 15,000p = 5,500$$

$$\boxed{p^* = \frac{5,500}{15,000} = \frac{11}{30} \approx 0.3667}.$$

Thus, for $p > \frac{11}{30}$, a risk-neutral agent prefers the Startup; for $p < \frac{11}{30}$, they prefer Bonds.

Q3. Expected utility with $U(W) = \sqrt{W}$

Bonds: certain wealth $W_B = 10,500$, utility

$$U_B = \sqrt{10,500} \approx \boxed{102.4695}.$$

Startup: expected utility

$$\mathbb{E}[U_S] = 0.5\sqrt{20,000} + 0.5\sqrt{5,000} = \frac{1}{2}(\sqrt{20,000} + \sqrt{5,000}).$$

Numerically,

$$\sqrt{20,000} \approx 141.4214, \quad \sqrt{5,000} \approx 70.7107,$$

so

$$\mathbb{E}[U_S] \approx \frac{1}{2}(141.4214 + 70.7107) = \boxed{106.0660}.$$

Comparison (utility):

$$\mathbb{E}[U_S] \approx 106.0660 > U_B \approx 102.4695 \implies \text{With } U(W) = \sqrt{W}, \text{ the startup is preferred.}$$

Certainty equivalents. The certainty equivalent (CE) is the risk-free amount of money that gives an individual the same amount of utility as a gamble or uncertain prospect. It represents the most a risk-averse person would pay to avoid a risk, or equivalently, the guaranteed amount they would accept instead of taking on the uncertainty. The difference between the expected value of a gamble and its certainty equivalent is the risk premium, which measures the amount of compensation an investor requires to take on risk.

The certainty equivalent (CE) is the sure amount with the same utility:

$$CE_B = (\sqrt{10,500})^2 = \boxed{10,500}, \quad CE_S = (\mathbb{E}[U_S])^2 = \left(\frac{\sqrt{20,000} + \sqrt{5,000}}{2} \right)^2 \approx \boxed{11,250}.$$

Since $CE_S > CE_B$, even a risk-averse agent with $U(W) = \sqrt{W}$ prefers the startup in this setup.

Summary

- Indifference bond rate: $\boxed{25\%}$.
- At $p = 0.4$, $\mathbb{E}[\text{startup}] = \$11,000 > \$10,500$ (risk-neutral prefers startup).
- Utility comparison with $U(W) = \sqrt{W}$: startup has higher expected utility and higher certainty equivalent.

Possible methods to solve these problems

1. Expected Value (EV)

$$EV = \sum_{i=1}^n p_i \cdot x_i$$

where p_i = probability of outcome i , x_i = payoff. **Rule:** choose the option with the highest EV.

2. Expected Utility (EU)

Introduce a utility function $U(W)$ to reflect risk preferences:

$$EU = \sum_{i=1}^n p_i \cdot U(x_i).$$

- Risk-neutral: $U(W) = W$.
- Risk-averse: $U(W) = \sqrt{W}$ or $\ln(W)$.
- Risk-seeking: $U(W) = W^2$.

Rule: choose the option with the highest EU.

3. Certainty equivalent (CE) & risk premium

The certainty equivalent is the guaranteed amount that yields the same utility as the risky option:

$$U(CE) = EU.$$

$$\text{Risk premium} = EV - CE.$$

4. Decision tree analysis

- Represent decisions (squares), chance events (circles), and payoffs (end nodes).
- Work backwards (*rollback analysis*): compute EV or EU at chance nodes, then decide at decision nodes.

5. Sensitivity analysis

Vary probabilities or payoffs to test robustness of the decision. Example: find the threshold probability where a risky option becomes less attractive than the safe option.

6. Simulation (Monte Carlo)

For complex problems:

- Generate many random outcomes.
- Approximate EV and risk profile.
- Useful in finance, healthcare, and project management.

7. Risk criteria / decision rules

- **Maximin (pessimistic):** choose the option with the best worst-case payoff.
- **Maximax (optimistic):** choose the option with the best best-case payoff.
- **minimax regret:** minimize the maximum regret (difference between chosen payoff and the best payoff in each state).

Summary

- Use **EV** when probabilities and payoffs are sufficient.
- Use **EU** or **CE** when risk preferences matter.
- Use **decision trees** for structured step-by-step analysis.
- Use **sensitivity analysis** and **simulation** for deeper insights.
- Consider **alternative criteria** for teaching different perspectives.

Few points to discuss further

Key concepts from systems thinking

- **Le Chateliers principle**: systems resist change; when perturbed, feedbacks push back toward equilibrium or generate new counter-forces.
- **John Galls systemantics**: “systems that work” evolve from smaller working systems; trying to design complex systems from scratch often fail.
- The system always **kicks back**: no matter how well-intended or designed, systems generate side effects, friction, delays, opposition.

Examples of systems pushback/failure:

- Healthcare.gov: built to simplify access, but technical failure locked out millions.
- Australias disability reforms: cost overruns beyond expectations.
- UKs contracts for difference policy: intended to speed up renewable energy rollout; ended up overstraining grid, creating long queues.

Principles for better decision making under risk

- Start with a simple working system.
Build something that does the core job; iterate and scales.
- Parallel experiments / small bets.
Try new approaches in limited scope alongside existing systems.
- Design for resistance & pushback.
Identify where opposition / friction will come; build in buffers, feedback channels.
- Avoid over-reliance on comprehensive top-down modeling.
Models are useful, but partial, often misleading. Dont assume full predictability.
- Continuous learning & adaptation.
Monitor outcomes, learn fast; iterate; be willing to abandon or pivot.

Traditional risk framework aligned with systems thinking

- **Risk identification**: identify feedback loops, unintended consequences, scale risks.
- **Risk assessment**: understand what parts of the system are unknown or changing; estimate uncertainty, not only likely paths.
- **Risk mitigation/control**: favor incremental roll-outs; modular design; safe fails; rollback paths.
- **Governance & accountability**: clear accountability lines, but also agility; reduce bureaucracy where it slows needed pivots.
- **Monitoring & early warning**: metrics for what matters; indicators of resistance, strain, failure.

Systems thinking (as in policy, engineering, management) assumes that if you can map out the parts and relationships of a system, you can understand, predict, and intervene. The focus is on external system structure and feedback loops. Failures arise because real-world systems are more resistant, dynamic, and unknowable than the models suggest.

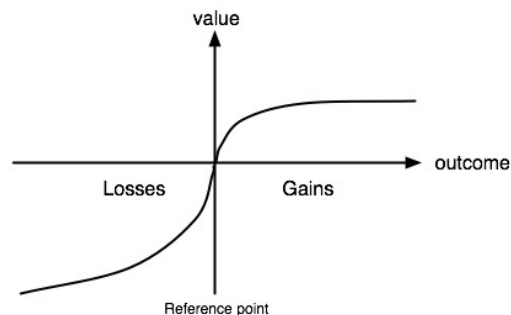
Cognitive Bayesian Models treat perception and action as probabilistic inference under uncertainty. The focus is not on mapping the entire system, but on how an agent updates beliefs in light of new evidence. Instead of assuming a complete map, Bayesian cognition accepts incomplete knowledge and emphasizes adaptation through updating priors.

Framing effects

- Decisions are heavily shaped by how information is presented.
- People tend to avoid risk when a choice is framed positively (gain frame), but take risks when its framed negatively (loss frame).
- Example: More people choose surgery when told “90% survival” compared to “10% mortality,” even though both are logically the same.
- Highlights that perception, not just facts, drives behavior.

Prospect theory and gainloss asymmetry (S-shaped value function)

- Developed by Kahneman & Tversky to explain deviations from rational economic behavior.
- Reference point: people evaluate outcomes compared to a baseline, not absolute wealth.
- Loss aversion: losses hurt about twice as much as equivalent gains feel good.
- The value function is S-shaped:
 - Concave for gains: people prefer a sure small gain over a gamble (risk-averse).
 - Convex for losses: people may gamble to avoid a sure loss (risk-seeking).
 - Explains phenomena like insurance buying (avoiding losses) and lottery playing (chasing unlikely gains).



Neuropsychology of risk aversion

- Brain regions involved:
 - Amygdala: signals fear and **potential threats**, increasing caution.
 - Insula: processes bodily signals of discomfort, linked to anticipating negative outcomes.
 - Prefrontal cortex: **weighs probabilities**, exerts rational control.
- Dopamine systems drive sensitivity to rewards, pushing toward risk in some contexts.
- Neuroimaging shows stronger amygdala/insula activity when people reject risky options, tying emotions directly to neural circuits.

Influence of emotion on decision making

- Emotions act as shortcuts, guiding quick decisions without detailed analysis.
- Fear/anxiety: more conservative, risk-averse choices.

- Anger/excitement: greater willingness to take risks and make snap judgments.
- Emotional memories (e.g., a past loss) can bias present decisions, even if the situation is different.
- Sometimes beneficial (avoids danger quickly), sometimes harmful (overreacting to small risks).

Cognitive control

- The mental ability to override automatic, emotional, or habitual responses.
- Centered in the prefrontal cortex and linked to working memory, inhibition, and planning.
- Allows balancing between emotion-driven impulses and rational analysis.
- Critical for delaying gratification, resisting biases, and making consistent long-term choices.
- Breakdowns in cognitive control (stress, fatigue, damage) increase reliance on emotional or heuristic-driven decisions.